Huawei Site-Specific SOP for Access Node JAY-ACC-13: Resolving Routing Protocol Flap Failure Due to BGP Route Churn, OSPF Adjacency Loss, or ISIS LSP Flooding

Node: JAY-ACC-13 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6624

Description

The routing protocol flap failure on Huawei's Access Node JAY-ACC-13, located at the Jayanagar Metro Site, can be attributed to the complex interplay between various routing protocols, namely BGP, OSPF, and ISIS. This failure domain can manifest in the form of excessive BGP route churn, OSPF adjacency loss, or ISIS LSP flooding due to misconfiguration, hardware issues, or environmental factors affecting the network's stability. Specifically, on Huawei equipment, this phenomenon can arise from the implementation of advanced routing features such as route reflectors, confederations, or multi-topology routing, which introduce additional complexity and potential points of failure. Moreover, the dynamic nature of routing protocols means that even small changes in the network topology or configuration can trigger a cascade of events, leading to a routing protocol flap. To mitigate this issue, it is crucial to identify and address the root cause of the problem, which may involve reviewing configuration files, analyzing log data, and performing diagnostic tests to isolate the affected protocols and determine the necessary corrective actions.

Pre-requisite Checks

- Ensure that the Jayanagar Metro Site is accessible and that all necessary personnel and equipment are present.

- Verify the status of redundant links and ensure that they are functioning correctly.

- Review the system logs and configuration files to identify any recent changes or potential causes of the issue.

Potential Root Causes

Cause	Probability
Misconfigured BGP route reflectors or confederations	High
Hardware issues affecting routing protocol processing	Medium
Environmental factors such as temperature fluctuations or power outages	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: The BGP routing
table will be displayed, showing the current routes and their
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attributes., Description: This command will help identify any issues with BGP route churn and allow us to analyze the routing table for any anomalies.

Command: display ospf routing-table, Expected Output: The OSPF routing table will be displayed, showing the current routes and their attributes., Description: This command will help identify any issues with OSPF adjacency loss and allow us to analyze the routing table for any anomalies.

Command: display isis routing-table, Expected Output: The ISIS routing table will be displayed, showing the current routes and their attributes., Description: This command will help identify any issues with ISIS LSP flooding and allow us to analyze the routing table for any anomalies.
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Resolution Steps

Step 1: Step: 1. Review BGP Configuration Files, Command: display bgp conf, Description: Review the BGP configuration files to ensure that they are correct and consistent with the current network topology.

Step 2: Step: 2. Check for Hardware Issues, Command: display hardware, Description: Use this command to check for any hardware issues that may be affecting routing protocol processing.

Step 3: Step: 3. Implement Redundancy Improvements, Command: config redundant-link, Description: Implement redundancy improvements to ensure that the network remains stable even in the event of a failure.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: The BGP routing table will be displayed, showing the current routes and their attributes., Description: Verify that the BGP routing table is stable and free from anomalies.

Command: display ospf routing-table, Expected Output: The OSPF routing table will be displayed, showing the current routes and their attributes., Description: Verify that the OSPF routing table is stable and free from anomalies.

Command: display isis routing-table, Expected Output: The ISIS routing table will be displayed, showing the current routes and their attributes., Description: Verify that the ISIS routing table is stable and free from anomalies.
```

Roll-back Procedures

- Step: 1. Restore Original Configuration Files, Command: config bgp conf original, Description: Restore the original BGP configuration files to ensure that the network returns to its previous stable state.

- Step: 2. Revert Hardware Changes, Command: config hardware revert, Description: Revert any hardware changes made during the troubleshooting process to ensure that the network returns to its previous stable state.